
ConnectSmart Research Lab

University of Tennessee at Chattanooga

Vahid Disfani, Ph.D.

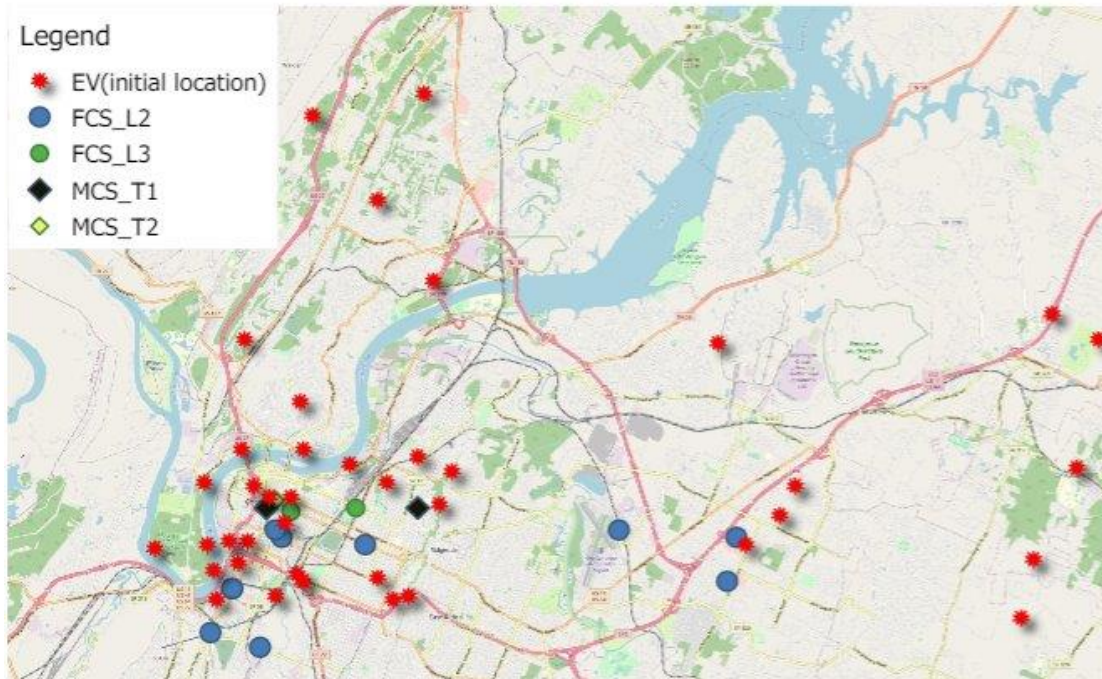
Vahid-disfani@utc.edu

Research Interests & Portfolio

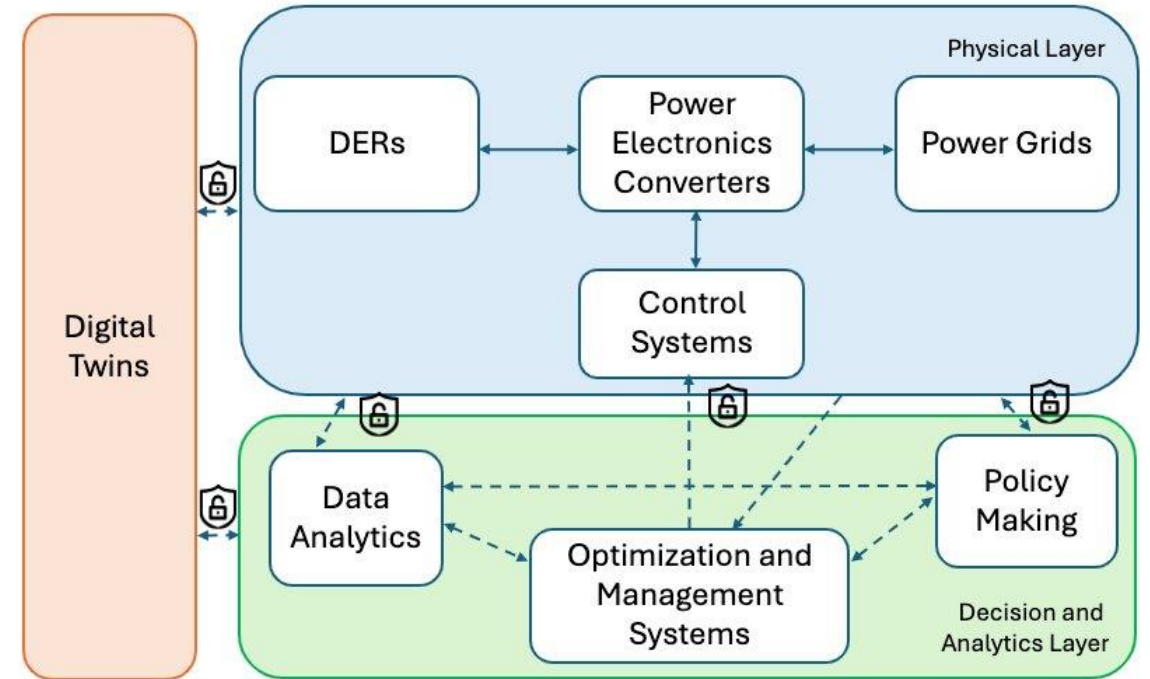
- Main Theme : Grid integration of Distributed Energy Resources
 - Energy management and optimization
 - Economic and Environmental Impacts
 - Power electronics and control
 - Electricity market participation
- Grants
 - As PI : \$1M Research Initiative Grant, funded by Tennessee Higher Education Commission (2025-2028)
 - Objective: establishing a new research center focused on smart energy systems
 - As co-PI : \$10M Urban EV Testbed in Chattanooga, funded by US Department of Transportation
- Team
 - One postdoc, 2 PhD students

Active Projects

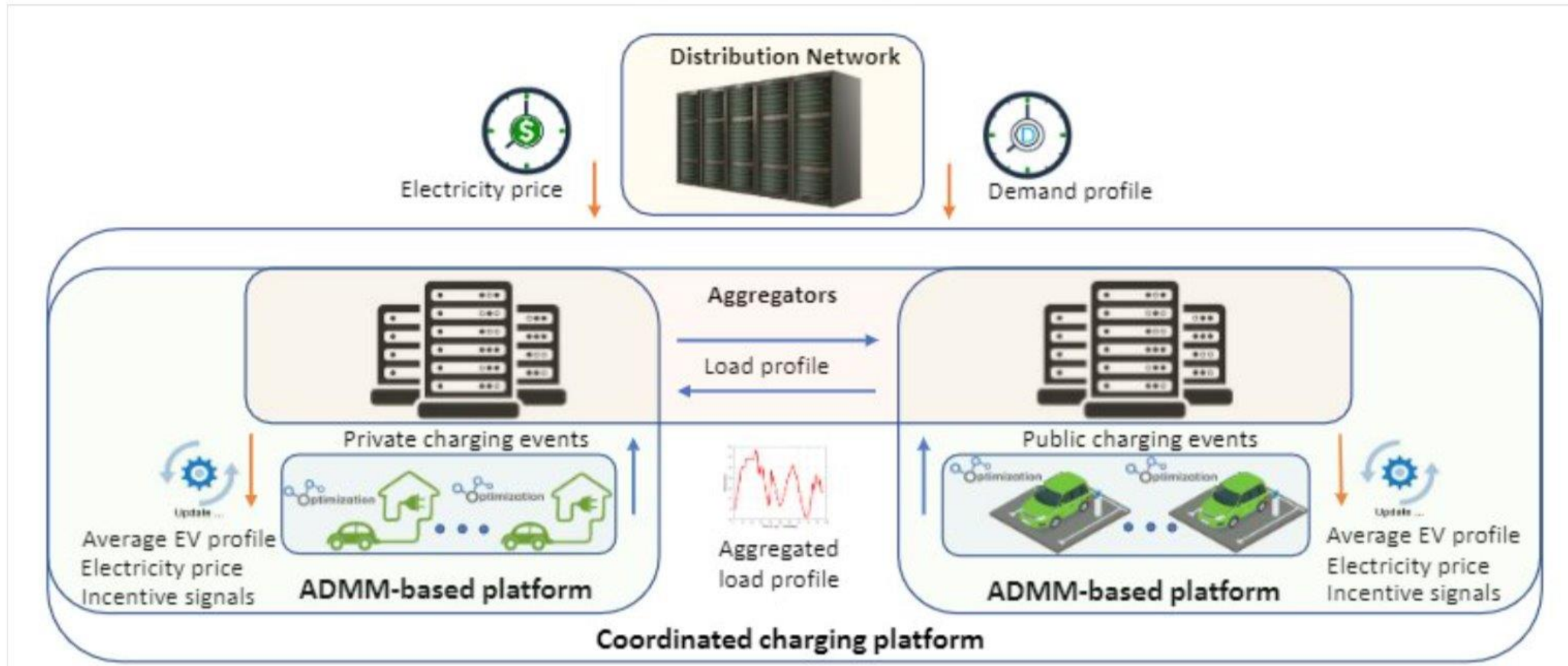
EV Charging Management



Smart Energy Systems Grant

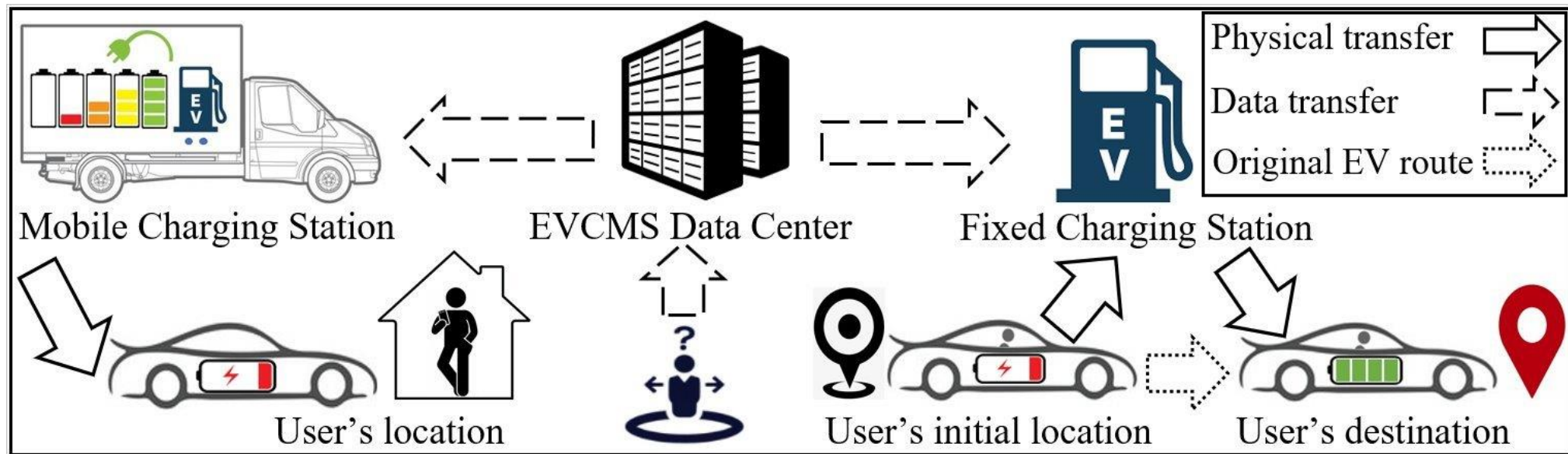


Distributed EV Charging Management



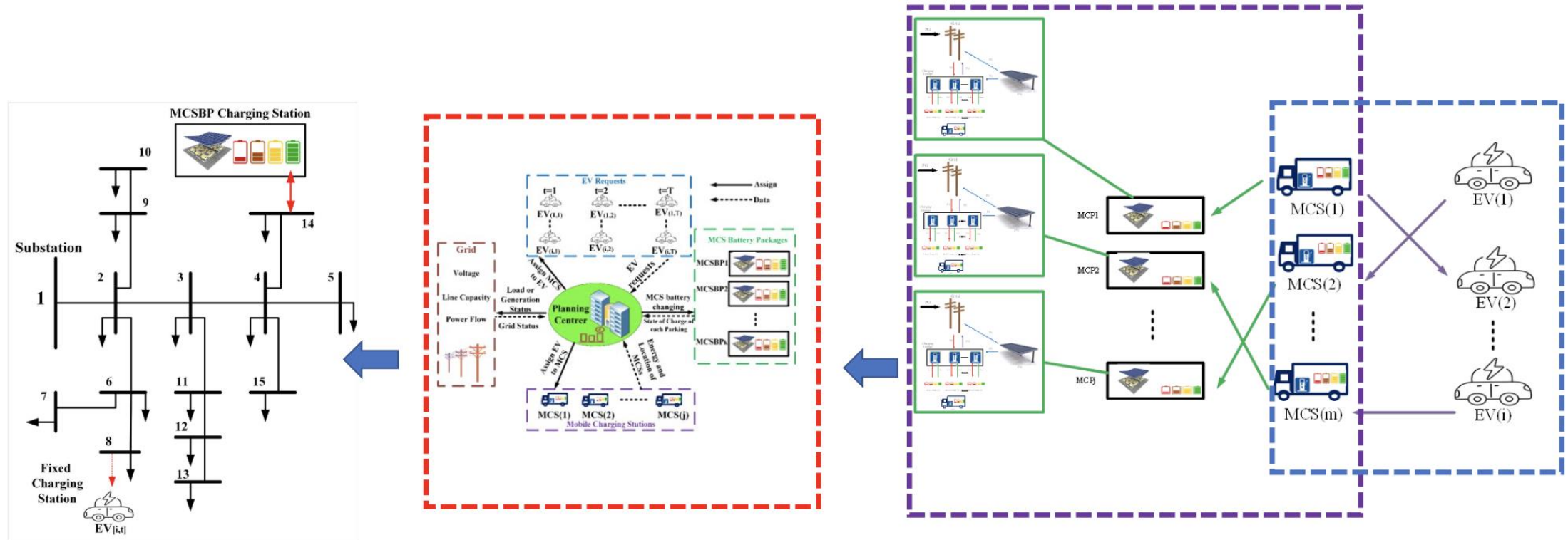
Afshar, et.al., "A Distributed Electric Vehicle Charging Scheduling Platform Considering Aggregators Coordination", IEEE Access, 2021.

Multi-Technology Charging Management



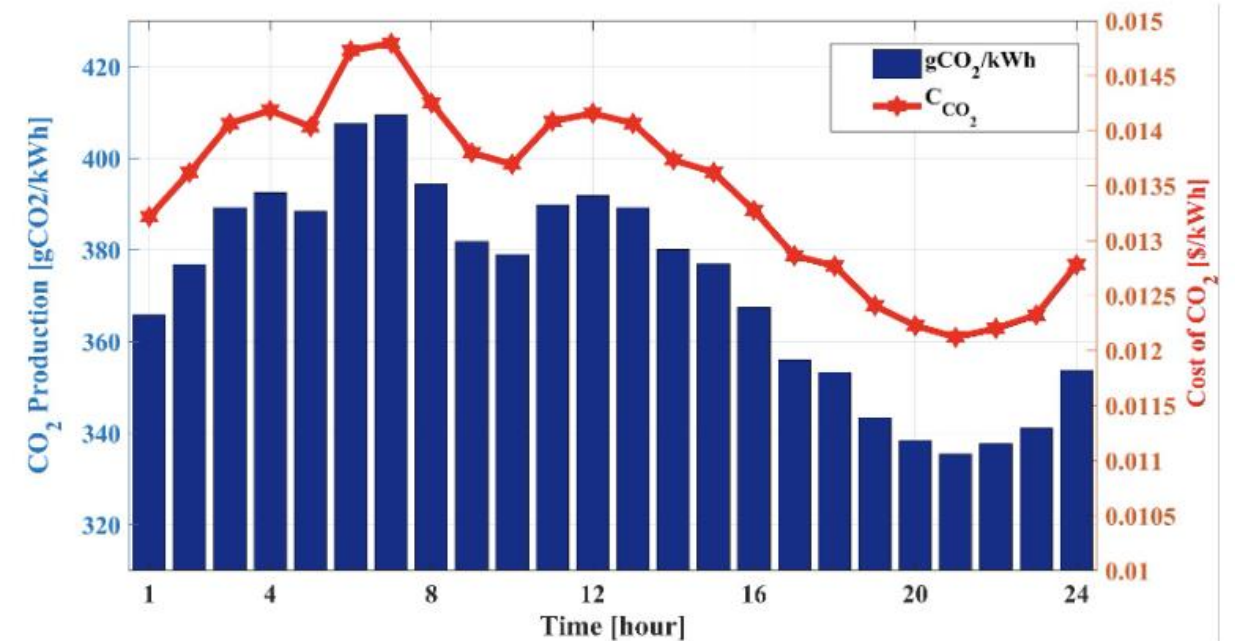
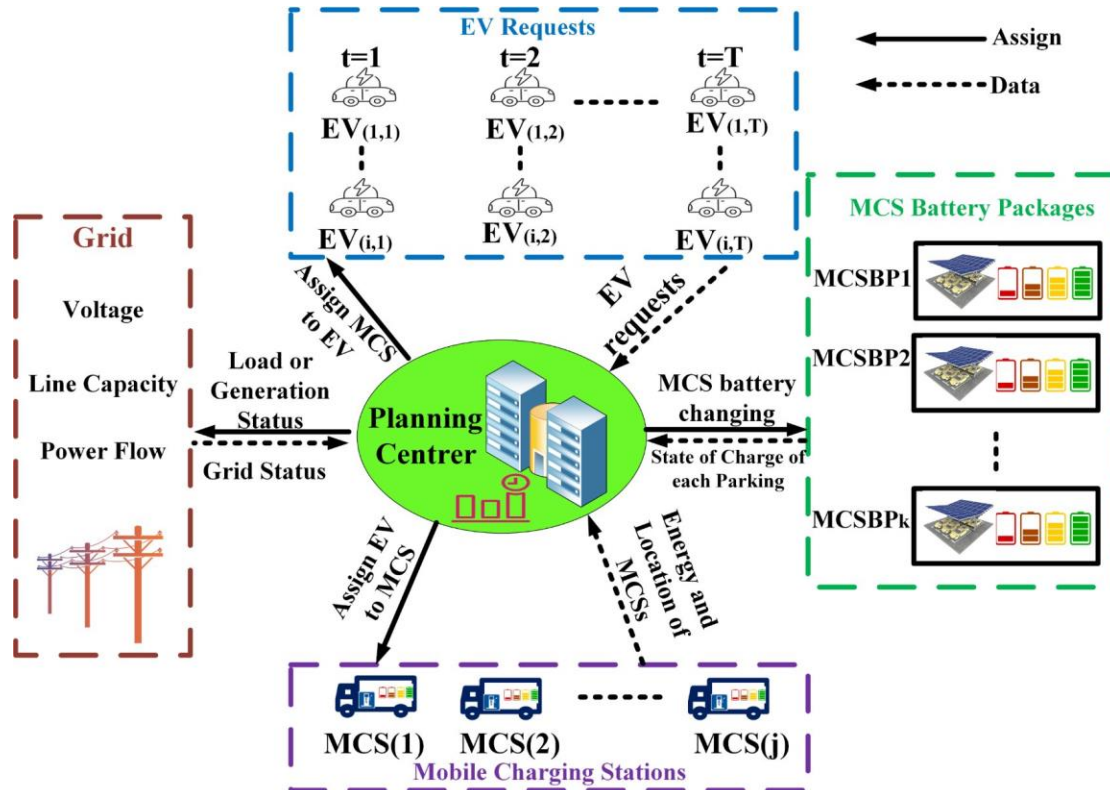
Afshar, et.al, "Mobile charging stations for EV charging management in urban areas: A case study in Chattanooga", Applied Energy, 2022.

Two-Stage Optimization for Mobile Charging Stations Management



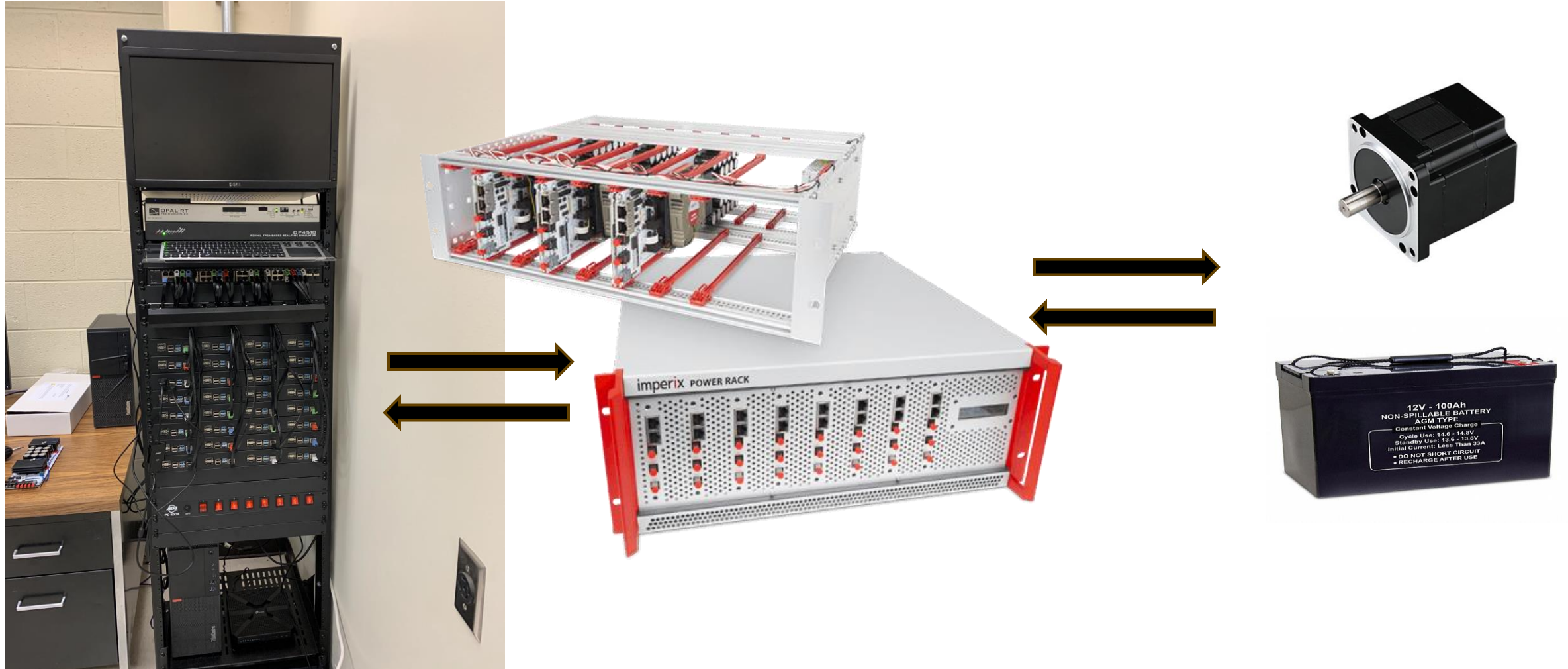
Alirezazadeh, et.al., "Deep reinforcement learning-based optimization of mobile charging station and battery recharging under grid constraints", *Energies*, 2025.

Environment-Aware EV Charging Management

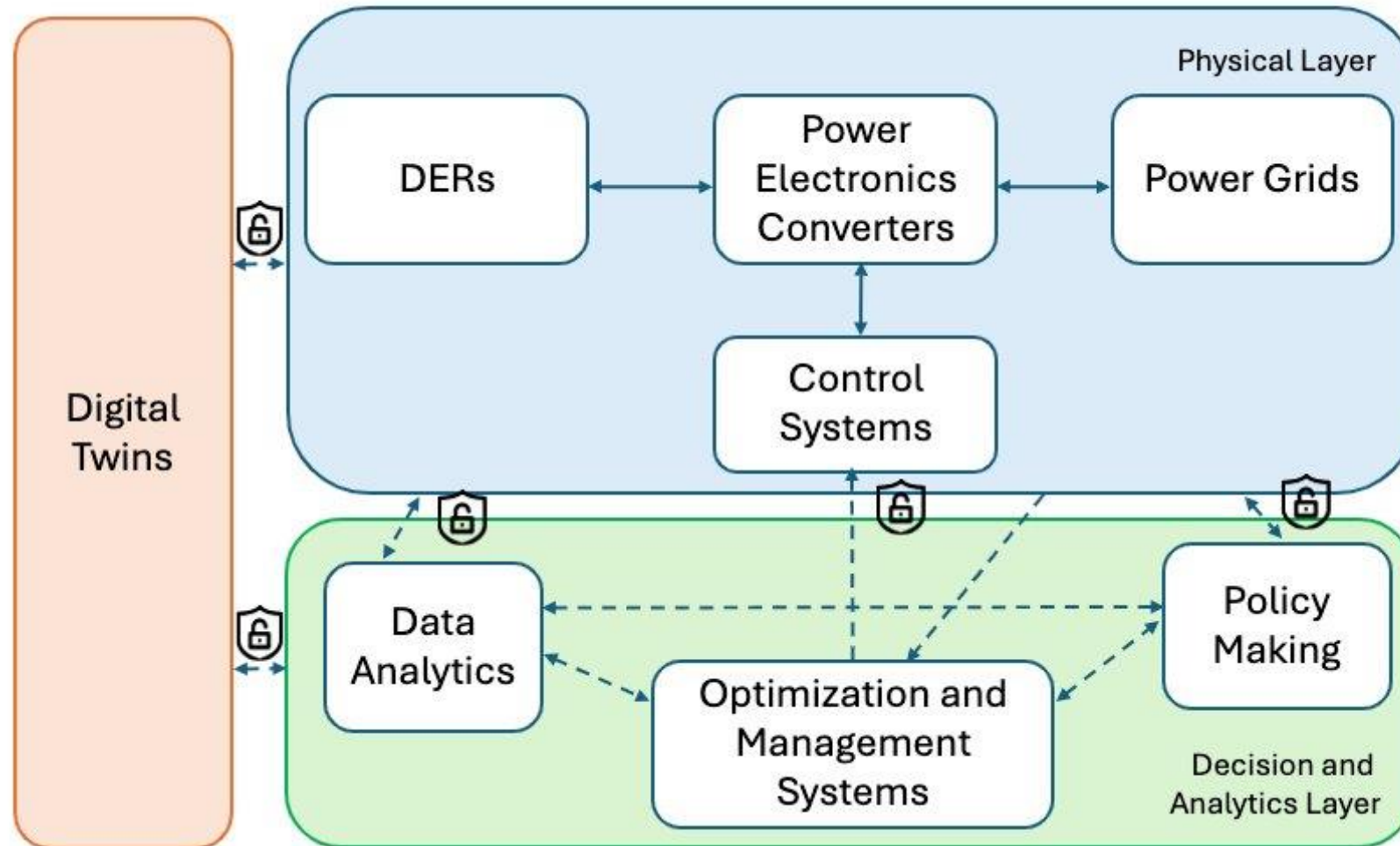


Alirezazadeh, et.al., "A new model of energy management for maximum social welfare and minimum carbon dioxide emissions considering parking sharing", Applied Energy, 2025.

Distributed Optimization and HIL Testbed



Smart Energy System Initiative Grant



Thrust Area 1 – Grid Integration, Power Electronics, and Control

DERs come with disadvantages

Power systems equipped with uncontrolled DERs inherit all adverse impacts of DERs, which eventually leads to failure of power systems in the long run

Effective control converts threats to opportunities

if equipped with advanced control and management systems, DERs will be invaluable assets to make power systems even more reliable and efficient

Core 1 : Power System Analysis

Advanced knowledge of power and energy systems is critical to identify and model the impacts of DERs on power systems and to develop technical and optimization solutions.

Core 2 : Power Electronics and Control

State-of-the-art PEC systems are required to seamlessly integrate DERs with power systems to gain the maximum benefits they offer and to mitigate their adverse impacts.

Thrust Area 2 – Energy Management and Optimization

Optimal Operation

Develop high-level energy management and optimization solutions to lead control systems toward best operations

Optimal Planning

Develop new planning and investment strategies for power systems with high penetration of DERs and microgrids

Multi-Agent Optimization

Develop multi-agent solutions to address private ownership of DERs and microgrids as well as data privacy and economic concerns

Thrust Area 3 – Data Analytics, AI, and Cybersecurity

01

Develop big data solutions to manage them and opens a unique opportunity for data analytics

02

Develop data-centered and AI-powered solution to unlock novel insights and approaches

03

Create distributed solutions for data analytics, often called federated learning algorithms, are required to address microgrids' privacy concerns

04

Investigate cybersecurity to secure communications among DERs and between DERs and power grids

Thrust Area 4 – Socioeconomics, Environment, and Policies

Social and economic Impact Studies

- Study and quantify socioeconomic and environmental impacts of grid integration of DERs and their clean energy
- Ensure equal access to DERs and their benefits by everyone in society, especially underrepresented communities

Environmental Impact Studies

- Promote clean energy
- Quantify emission savings
- Investigate novel environmental challenges
- Study impacts of DER manufacturing processes
- Study DER disposal and recycling impacts

Policy and Regulations

- Study impacts of policies and regulations in investment strategies
- Develop effective policies and regulations